

HAT-P-18b

R-2173

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-18_220419 · created 2026-08-13T00:47:58+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2459688.77243 +/- 0.00043 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.182 +/- 0.054
Transit depth (Rp/Rs)^2	3.3 +/- 1.95 [%]
Orbital Inclination (inc)	87.46 +/- 1.86
Airmass coefficient 1 (a1)	1.2 +/- 0.65
Airmass coefficient 2 (a2)	-0.087 +/- 0.088
Scatter in the residuals of the lightcurve fit is	51.74 %
Best Comparison Star	None
Optimal Aperture	2.29
Optimal Annulus	3.0
Transit Duration (day)	0.062 +/- 0.033

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.182 ± 0.054 vs 0.1356 (deviation 0.86σ)
Duration fit vs published	0.062 d vs 0.1129 d (deviation 1.54σ)
Mid-time O–C	-1.3 min
Depth SNR	3.4
Residual scatter	51.74 %

Light curve

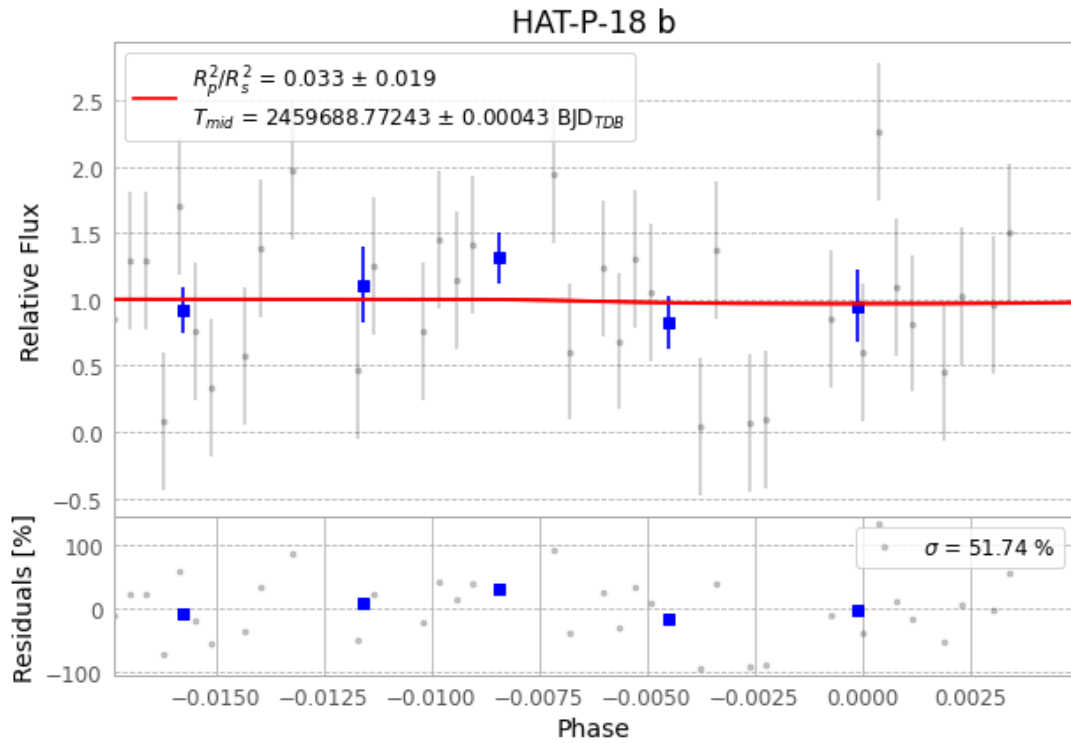


Figure 1 — FinalLightCurve_HAT-P-18 b_2022-04-18.png (SHA-256 8ffc91c266f56689...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [421, 139], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 07b2b3c5abb207d8...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

63 FITS frames (41 MB), HATP-18220419040913.FITS → HATP-18220419071514.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-18-220419.log (f2b0b017a021...), FinalLightCurve_HAT-P-18 b_2022-04-18.png (8ffc91c266f5...), FinalLightCurve_HAT-P-18 b_2022-04-18.csv (79e3362bc8a7...)

Compute resources

Wall time 179.8 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-13 00:47Z.